IIoT and Cloud Computing (CS5005)

Lecture – 32

Resource Management

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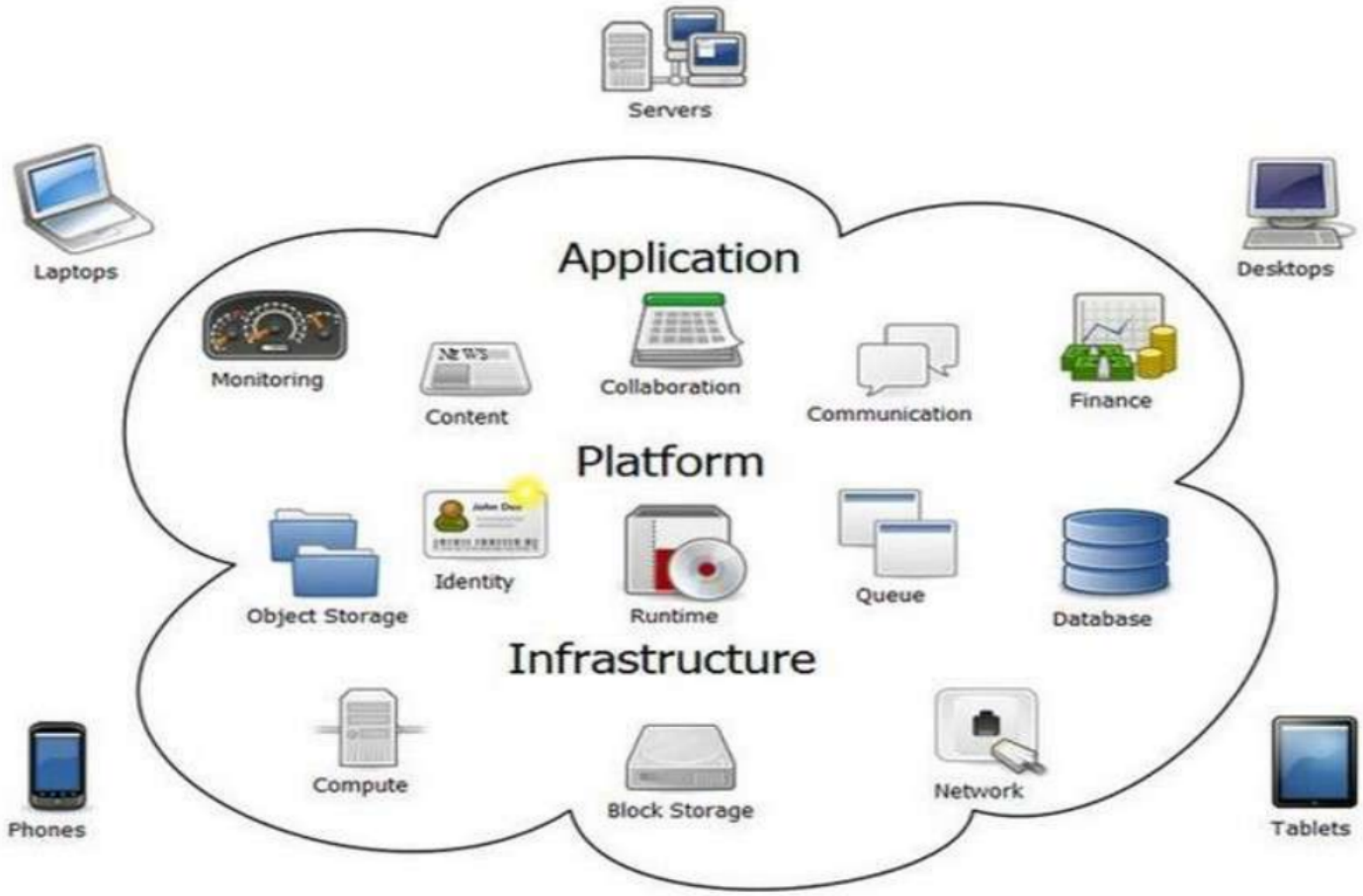
Different Resources in Computing



Source: http://www.cse.hcmut.edu.vn/~ptvu/gc/2012/GC-pp.pdf

Resources types

- **Physical resource**
  - Computer, disk, database, network, scientific instruments.
- **Logical resource**
  - Execution, monitoring, communicate application.

Source: http://www.cse.hcmut.edu.vn/~ptvu/gc/2012/GC-pp.pdf

Resources Management

- The operations used to control how capabilities provided by Cloud resources and services can be made available to other entities, whether users, applications, services in an efficient manner.

Source: http://www.cse.hcmut.edu.vn/~ptvu/gc/2012/GC-pp.pdf

Data Center Power Consumption

- Currently it is estimated that servers consume 0.5% of the world's total electricity usage.
- Server energy demand doubles every 5-6 years.
- This results in large amounts of CO_2 produced by burning fossil fuels.
- Need to reduce the energy used with minimal performance impact.

Ref: Efficient Resource Management for Cloud Computing Environments, by Andrew J. Younge, Gregor von Laszewski, Lizhe Wang, Sonia Lopez-Alarcon, Warren Carithers,